

Practical no. 15

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Class :- CO3K – B

Subject :- DSU

Program :-

```
#include<stdio.h>

#include<stdlib.h>

#include<conio.h>

struct Node{

int coefficient;

int exponent;

struct Node*next;

};

struct Node*createNode(int coeff,int exp){

struct Node*newNode=(struct Node*)malloc(sizeof(struct Node));

newNode->coefficient=coeff;

newNode->exponent=exp;

newNode->next=NULL;

return newNode;

}

struct Node*insertTerm(struct Node*head,int coeff,int exp){

struct Node*current=head;

struct Node*newNode=createNode(coeff,exp);

if(head==NULL || exp>head->exponent){

newNode->next=head;

return newNode;

}

while(current->next!=NULL&&exp<current->next->exponent){
```

```

current=current->next;

}

newNode->next=current->next;

current->next=newNode;

return head;

}

struct Node*addpoly(struct Node*poly1,struct Node*poly2){

struct Node*result=NULL;

while(poly1!=NULL&&poly2!=NULL){

if(poly1->exponent==poly2->exponent){

result=insertTerm(result,poly1->coefficient+poly2->coefficient,poly1->exponent);

poly1=poly1->next;

poly2=poly2->next;

}

else if(poly1->exponent>poly2->exponent){

result=insertTerm(result,poly1->coefficient,poly1->exponent);

poly1=poly1->next;

}

else{

result=insertTerm(result,poly2->coefficient,poly2->exponent);

poly2=poly2->next;

}

}

while(poly1!=NULL){

result=insertTerm(result,poly1->coefficient,poly1->exponent);

poly1=poly1->next;

}

```

```

while(poly2!=NULL){
    result=insertTerm(result,poly2->coefficient,poly2->exponent);
    poly2=poly2->next;
}

return result;
}

void display(struct Node*head){
    struct Node*current=head;

    if(head==NULL){
        printf("\nList is empty!!");
    }

    while(current!=NULL){
        printf("%dx^%d",current->coefficient,current->exponent);

        current=current->next;

        if(current!=NULL){
            printf("+");
        }

        printf("\n");
    }

    void main(){

        struct Node*poly1=NULL;

        struct Node*poly2=NULL;

        struct Node*head;

        struct Node*sum=NULL;

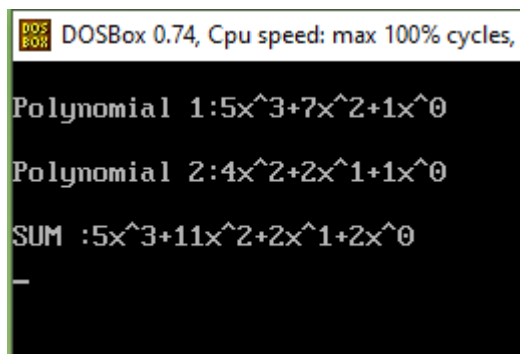
        poly1=insertTerm(poly1,5,3);

        poly1=insertTerm(poly1,7,2);

```

```
poly1=insertTerm(poly1,1,0);  
poly2=insertTerm(poly2,4,2);  
poly2=insertTerm(poly2,2,1);  
poly2=insertTerm(poly2,1,0);  
clrscr();  
printf("\nPolynomial 1:");  
display(poly1);  
printf("\nPolynomial 2:");  
display(poly2);  
sum=addpoly(poly1,poly2);  
printf("\nSUM :");  
display(sum);  
getch();  
}
```

OUTPUT :-



DOSBox 0.74, Cpu speed: max 100% cycles,

```
Polynomial 1:5x^3+7x^2+1x^0  
Polynomial 2:4x^2+2x^1+1x^0  
SUM :5x^3+11x^2+2x^1+2x^0  
-
```